RIN RAY

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Mathematician and computational scientist · PhD, Northwestern · Universität Münster

PROFILE

I work on what can be learned from the traces systems leave behind. Over more than a decade, my work has crossed computational physics, medical technology and assistive robotics, biomedical sensing, and machine learning on biological signals, building models and instruments for working with partial and indirect information. I came to mathematics from the laboratory and never really left; I am now a postdoctoral fellow in homotopy theory and arithmetic geometry, alongside active research in medicine: pharmacology and drug repurposing, chronic pain and sensory processing, and the gut–brain axis, built on close reading of the clinical and preclinical literature.

EDUCATION

2017–2023 **Northwestern**; PhD in Mathematics; advised by Paul Goerss

- *A Geometric Model of Higher K-theories at Height $h=p^{k-1}(p-1)$ via Families of Ramified Curves*

2015–2017 **UChicago**; Divisional Masters Program in Mathematics; advised by Peter May and Kazuya Kato

- *Calculating $\pi_*(tmf)$ at the prime 2*
- *Models of Formal Group Laws of Every Height*

2012–2013 **George Mason University**; B.S. in Computational Physics (graduated winter 2013, at 16); advised by Dimitrios Papaconstantopoulos

- *Contextual Machine Learning through the Analysis and Chunking of Partially Translated Grade 2 Braille*
- *Predicting Superconductivity Transition Temperature via the Augmented Plane Wave Model*

2010–2012 **Mary Baldwin College**; Program for the Exceptionally Gifted: Mathematics, Computer Science, and Physics; advised by John Ong

POSITIONS

2023–Present **CRC Postdoctoral Fellow**; Universität Münster; Arithmetic and Homotopy Theory Working Group, with Thomas Nikolaus and Christopher Deninger. Arithmetic patterns in homotopy theory and physics: geometric models of the Lubin–Tate action via moduli stacks of curves, and zeta functions in homotopy theory.

- chromatic homotopy theory · moduli stacks of curves · arithmetic geometry

2015 **Visiting Researcher**; Santa Fe Institute of Complex Systems; Santa Fe, NM

2015 **Visiting Researcher**; Max-Planck-Institut für Mathematik; Bonn, Germany

2014–2016 **Thiel Fellow**; Thiel Foundation; Berkeley, CA; medical and assistive technology

2013–2015 **Bioacoustics Engineer**; Vium; Belmont, CA

2013 **Software Engineering Intern**; Cloudera; Palo Alto, CA; Hadoop · Hive · large-scale data processing

2013 **Computational Chemistry Researcher**; George Mason University; Fairfax, VA; modeling the superconducting properties of intermetallic compounds, with D. Papaconstantopoulos

2013 **Computational Semantics and Machine Learning Researcher**; George Mason University; Fairfax, VA; language processing, and vision for aerial robots

2012 **Human–Computer Interaction Intern (Autonomous Robotics)**; George Washington University; Washington, DC

2011 **Research Assistant, Materials Science Lab**; Mary Baldwin College; Staunton, VA; memristors

RESEARCH

MEDICINE AND MEDICAL TECHNOLOGY

Hidden states, read from the traces a body emits, and instruments built for bodies.

2026 CURRENT **One Fault, Many Names: No Cohort Sees the Whole Vector**; solo; follow-up to *Biome of Overstimulation*; proposal to measure habituation as a failure-to-decrement slope on several channels (skin, gut, cortex) in the same people, using instruments clinics already own run in repeated mode. Each person's deficit splits into severity, what the conditions share, and direction, where the fault lands; the hypothesis is falsified unless directions form a few tight clusters.

Proposal; untested.

habituation · quantitative sensory testing · visceral sensitivity (barostat) · evoked responses · comorbidity · latent-variable models · directional statistics (von Mises–Fisher mixtures)

2026 CURRENT — **Neostigmine in feline idiopathic megacolon**; owner-reported case study of an acetylcholinesterase inhibitor used off-label to restore colonic motility, documented to a standard usable in veterinary consultation. A treatment that worked and had not been written down.

veterinary pharmacology · enteric motility and cholinergic transmission · acetylcholinesterase inhibition · case documentation of an unpublished treatment

2026 CURRENT — **Memantine as a candidate non-opioid analgesic in central sensitization**; review of the mechanistic rationale, comparative pharmacology against ketamine and dextromethorphan, and the clinical trial record for memantine in chronic pain, with a proposed design for a publicly funded trial. Evidence synthesis only; the pooled analysis of eleven trials is not significant.

pharmacology · NMDA receptor antagonism · central sensitization · opioid tolerance and hyperalgesia · drug repurposing · trial design · health policy

2024– CURRENT — **Biome of Overstimulation: Sensory Sensitivity as a Unifying Mechanism**; with L. Estinto; synthesis arguing that failure of habituation is a shared upstream mechanism, drawing on connectivity imaging, glutamatergic excitability, microbiome composition and enteric regulation of immunoglobulin secretion. Hypothesis-generating; not experimentally validated.

systems biology · neuroimmunology · sensory processing and habituation · gut microbiome · autoimmunity · chronic pain · glutamatergic excitability · evidence synthesis

2014 — **Decoder convergence for intracortical motor prostheses**; convergence analysis of Kalman-style decoders under signal drift: how fast the filter settles, how far it degrades as glial scarring thins the recorded population, and how much adaptive recalibration recovers. Then optical recording via fluorescent calcium indicators as an alternative to penetrating electrodes.

intracortical brain–computer interfaces · decoder convergence analysis · adaptive filtering under non-stationary signals · foreign body response and chronic signal loss · optical recording

2013–2015 — **Unsupervised classification of rodent ultrasonic vocalizations**; Vium; quasi-real-time clustering of calls from continuously monitored cages, deployed as a non-invasive preclinical readout for compound effects on stress and libido. Run continuously, not on a corpus after the fact, several years before MUPET and DeepSqueak made unsupervised clustering standard. Found a previously uncharacterized female–female vocalization. Internal reports: *A New Female–Female Mouse Vocalization Discovered via Unlabeled Machine Learning*; *On the Detection and Prevention of Aggression in Lab Mice via Quasi-Real Time Analysis*.

computational bioacoustics · unsupervised clustering of ultrasonic vocalizations (USVs) · quasi-real-time audio processing · affective-state readout · preclinical safety pharmacology · 3Rs refinement

2013 — **Remote photoplethysmography on a head-mounted display**; HackMIT; cardiac pulse recovered from video without contact. Withdrawn on ethical grounds: the measured party cannot detect or refuse the measurement.

remote photoplethysmography from video · contactless physiological sensing · computer vision · consent and surveillance ethics

2013–2015 — **Modular robotic retrofits for powered wheelchairs**; four assemblies that bolt onto a chair already fitted to its user: automated pressure redistribution, powered seat elevation, assisted transfer, rough-terrain drive. Requirements elicited from wheelchair users with spinal cord injury. Written as a nonprovisional patent application, deliberately not filed, and released under Creative Commons.

assistive robotics · modular retrofit design · soft robotics · pressure injury prevention · rehabilitation engineering · user-led requirements · gaze-controlled mobility (mentorship)

2011–2014 — **Portable detection of gluten and allergenic proteins**; a laser and avalanche-photodiode optical assay, abandoned for insufficient signal-to-noise against protein-rich matrices; then a G12 antibody colorimetric assay. The limit was antibody cost per assay and sampling variance in food, not sensitivity.

point-of-consumption allergen detection · laser absorption spectroscopy · G12 antibody colorimetric immunoassay · assay cost and sampling limits

BODIES IN MOTION

A body reading the traces of its own movement.

2012 — **Graceful degradation under sensor failure: manipulation on proprioception alone**; George Washington University; with vision and all external feedback disabled, a Willow Garage PR2 learned a shape-matched insertion task via reinforcement learning from commanded joint positions and gripper sensing only. Also two-arm load equalization for objects with off-center mass.

graceful degradation · proprioception-only contact-rich manipulation · blind peg-in-hole insertion · reinforcement learning (policy search) · two-arm load equalization · human–robot interaction

2012 — **Monocular visual SLAM with motion planning on a micro aerial vehicle**; George Mason University; PTAM-class SLAM processed off-board, metric scale recovered by fusing sonar altimetry and inertial measurement, coupled to frontier-based planning on a partially built map.

simultaneous localization and mapping · monocular visual SLAM with parallax bootstrapping · loop closure · sonar and inertial sensor fusion · motion planning on a partial map · unmanned aerial vehicles

2011 — **Gait-based re-identification for robot person following**; hexapedal platform; target enrolled by proximity and re-identified by gait, not appearance.

robot person following · gait-based re-identification · computer vision · statically stable hexapedal locomotion

MATTER

Behavior inferred from the traces that structure leaves.

2015 **Dependent and directed type systems for multiphysics models**; Santa Fe Institute; proposal that multiphysics models be written in a dependent type language: classical fields are sections of fiber bundles and dependent types are fibrations, so model structure, including direction of coupling, can be carried in the type and exploited by the compiler. Report: *Simplifying Multiscale Modeling* (Santa Fe Institute).

multiscale and multiphysics modeling · dependent and directed type systems · compositional model building · category theory · scientific computing

2012–2013 **Electronic-structure prediction of superconducting transition temperature**; George Mason University, advised by D. Papaconstantopoulos; augmented plane wave band structure feeding an electron–phonon coupling estimate and a McMillan/Allen–Dynes transition temperature for intermetallic compounds. Undergraduate thesis: *Predicting Superconductivity Transition Temperature via the Augmented Plane Wave Model*.

electronic-structure calculation · augmented plane wave method · electron–phonon coupling · superconducting transition temperature of intermetallic compounds · computational materials screening

2012 **Standing-wave mode structure of an RF discharge in a polyhedral cavity**; 2.45 GHz argon discharge in an evacuated aluminum polyhedron; the cavity geometry selects the supported resonant modes. Benchtop analogue for deployable conductive structures in ionospheric plasma.

RF plasma · standing-wave mode structure · cavity resonance · electromagnetics · deployable structures in ionospheric plasma

2012 **Flexible TiO_2 resistive-switching devices**; Mary Baldwin College; fabrication and bend-cycle characterization of memristive devices on compliant substrates, toward colocated storage and computation for body-worn sensing.

memristive devices · flexible TiO_2 resistive switching · thin-film fabrication · bend-cycle characterization · in-memory computing

FRAGMENTS

Structure recovered from the traces a language leaves behind.

2013 **Unsupervised grammar and lexicon induction from partially parallel text**; George Mason University; probabilistic lexicon induction over weakly parallel corpora, evaluated on contracted Grade 2 Braille as a test language with known ground truth; generalizes to incompletely deciphered scripts. Undergraduate thesis: *Contextual Machine Learning through the Analysis and Chunking of Partially Translated Grade 2 Braille*.

natural language processing · unsupervised grammar and probabilistic lexicon induction · partially parallel corpora · decipherment · contracted Braille · accessibility

2013 **Abstractive summarization of biomedical literature for clinical readers**; George Mason University; automated computational semantics aimed at reducing a paper to a triage-sufficient summary that keeps its evidence intact.

natural language processing · automated computational semantics · abstractive summarization · biomedical text · evidence fidelity

SELECTED MATHEMATICS PAPERS

FULL LIST AT RIN.IO/ABOUT

Sept 2026 ***On the Chromatic Splitting Conjecture in Coheight 1***; joint with T. Barthel, L. Mann, A. Senger, T. Schlank, J. Weinstein, X. Zhou; preprint

Sept 2025 ***Tiny and Unique: The Lubin–Tate Action at Height p−1 from the Minimal Curve with Its Symmetries***; arXiv:2509.23428, v3 Oct 2026; submitted to *AG&T*

July 2025 ***Towards the p = 3 Kervaire Invariant Problem: The E_2-page for the Homotopy Fixed Points Spectral Sequence Computing $\pi_*(E_6^{hC_9})$***; joint with E. Belmont; arXiv:2507.10157; submitted

July 2025 ***Modeling Group Actions on Stacks (Especially the Lubin–Tate Action)***; arXiv:2507.00309, v2 Oct 2026; submitted to *Journal of Geometry and Physics*

2022 ***Automorphisms of Abelian Varieties and Principal Polarizations***; joint with D. Lee; Rendiconti del Circolo Matematico di Palermo, Series 2, vol. 71, pp. 483–494; with a code base written for high-performance computing

2020 ***Towards Directed Collapsibility***; joint with R. Belton, R. Brooks, S. Ebli, L. Fajstrup, B. T. Fasy, N. Sanderson, E. Vidaurre; Advances in Mathematical Sciences, vol. 21, pp. 255–271

FELLOWSHIPS AND AWARDS

2022 **Gelfand Prize**, Northwestern University; for outstanding progress in research and extraordinary contributions to the research goals of the graduate community

2022 **Nominee, Presidential Fellowship**, Northwestern University

2017 **NSF Graduate Research Fellowship**

2014–2016 **Thiel Fellowship** ($100K); awarded at 17; mentor Edward Frenkel

2013	**Outstanding Rising Senior Award** in Computational and Data Sciences, SPACS

METHODS

Medicine	pharmacology and mechanism of action · drug repurposing · clinical trial appraisal and design · evidence synthesis and pooled analysis · neurophysiology of pain and habituation (QST, conditioned pain modulation, evoked responses) · enteric motility · rehabilitation engineering
Inference	unsupervised learning and clustering · probabilistic lexicon induction · adaptive filtering and decoder analysis · policy search · evidence synthesis
Signals	real-time audio processing · neural signal decoding · computer vision · visual SLAM · sensor fusion · optical and immunoassay measurement
Modeling	electronic-structure calculation · multiscale and multiphysics modeling · type-theoretic model structure · large-scale data processing (Hadoop, Hive) · high-performance computing
Mathematics	homotopy theory · arithmetic geometry · moduli theory · category theory and type theory
Software	Python · NumPy · SciPy · scikit-learn · Git · Linux/Unix

TEACHING

2025	**Mathe trifft Kunst** (Math Meets Art), Day of Mathematics, Mathematics Münster Cluster of Excellence; organized all the children's activities, about 100 children: Escher and wallpaper groups, dual polytopes, Sierpiński triangles, composition in photography
2024	**Feedback seminar**, Homotopy working group, Universität Münster; giving and receiving feedback on talks, with assigned feedback-givers for every speaker
2018–2025	**Speaker coaching**; coached every speaker in the seminars I organized, at Northwestern and in Münster
2017–2023	**Teaching Assistant**, Northwestern; honors linear algebra and multivariable calculus, linear algebra, discrete mathematics · rated 5.65/6 for interest in teaching
2020–2023	**Directed Reading Program**, Northwestern; weekly one-on-one reading with an undergraduate, ending in a talk to their peers
2015–2017	**Research Experience for Undergraduates**, University of Chicago; weekly lectures, and guiding each student through an individual research project
2016–2017	**Math Circles**, South Side of Chicago; weekly mathematics with children aged 6–12
2012	**Teaching Assistant, College Algebra**, Mary Baldwin College

COMMUNITY AND OUTREACH

2024	**Organizer, Young Topologists Meeting**, Münster; drew the logo, designed the schedule with N. Pacheco-Tallaj
2024	**Queer in Math**; panelist at Queer in Math Day, MPI Leipzig
2023	**Set up a refugee program for PhD students**, Universität Münster; hosted PhD students for up to six months, waiving the usual processing time so they could leave at once
2022	**Advocate for Ukrainian students**, Northwestern; won incompletes and leniency for students who had to stay registered full-time to keep their visas
2021–2023	**Causeway Bridge Program** and **TGS Diversity Peer Mentorship**, Northwestern; mentoring students from diverse backgrounds into and through graduate school
2021–	**Founder, Mental Health Math Buddy Program**, Northwestern; pairs first- and second-year graduate students with older students for quarterly check-ins; ongoing
2017–2023	**Building Cities with Young Girls**, Evanston; weekly forts and cityscapes with young girls, to give working mothers in math time back
2022	**Founder and Treasurer, Graduate Life Appreciation Fund**, Northwestern; helps graduate students take part in activities they could not otherwise afford
2020–2023	**Medical Foster**, Evanston Animal Shelter and Red Door Animal Shelter, Evanston; Tree House Humane Society, Chicago; special-needs adult cats and neonatal kittens, too fragile for the shelter
ongoing	***Good Fibrations***, math blog with drawings; and *re-search*, an interactive story
2014–	**Founding contributor, Topos House**, South San Francisco; mathematicians, biologists and artists; ongoing